Retail Sales Forecasting Using Machine Learning

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I. INTRODUCTION

Retail sales forecasting helps businesses predict customer demand, manage inventory, and improve operational planning. However, sales are influenced by seasonal trends, holidays, promotional activities, and changing economic conditions, making accurate predictions challenging. Machine learning provides opportunities to analyze historical data, identify patterns, and improve forecasting accuracy. This research focuses on developing and evaluating machine learning models using the Walmart Store Sales Forecasting dataset from Kaggle. By comparing traditional statistical methods with machine learning techniques, the study aims to identify a reliable forecasting approach that supports better retail business decisions and effective resource management.

II. PROBLEM STATEMENT

Retail businesses face challenges in predicting future sales due to seasonal demand, holidays, promotional activities, and changing economic conditions. Inaccurate forecasts can result in excess inventory, product shortages, and increased operational costs. Traditional forecasting methods may struggle to capture complex relationships among factors influencing sales. Machine learning techniques offer opportunities to analyze historical data and improve prediction accuracy. However, their effectiveness compared with traditional approaches needs further investigation. Therefore, this research aims to evaluate selected machine learning models

using the Walmart Store Sales Forecasting dataset and identify the most reliable approach for predicting future retail sales and supporting effective business planning.

III. RESEARCH AIM AND OBJECTIVES

The primary aim of this research is to develop and evaluate machine learning-based models for predicting future retail sales using the Walmart Store Sales Forecasting dataset. The study will compare traditional statistical forecasting methods with machine learning techniques to identify an accurate and reliable approach for retail sales prediction. The research objectives are to analyze historical Walmart sales data and identify trends, seasonal variations, and holiday-related sales patterns; clean, preprocess, and transform the dataset to improve data quality; identify important variables influencing retail sales, including holidays, promotional activities, and economic indicators; and develop forecasting models using Seasonal-Naive, statistical time-series, Random Forest, and Gradient Boosting techniques. Additionally, the study will evaluate and compare model performance using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Weighted Mean Absolute Error (WMAE) to identify the most suitable forecasting model and provide recommendations for improving retail business planning.

IV. RESEARCH QUESTIONS

The project will address the following research questions:

1. What historical sales patterns and seasonal trends can be identified in the Walmart Store Sales Forecasting dataset?
2. Which variables, including holidays, promotional markdowns, store characteristics, and economic indicators, have the greatest influence on retail sales forecasting?
3. How does the forecasting performance of machine learning models compare with traditional statistical forecasting methods?

4. Which selected forecasting model provides the lowest prediction error when evaluated on previously unseen sales data?
5. How can the forecasting results support retail business decisions related to inventory management, staffing, and promotional planning?

V. LITERATURE REVIEW

The following literature review examines previous research on retail sales forecasting, focusing on traditional statistical methods, machine learning algorithms, and advanced predictive techniques. It highlights key findings from existing studies and establishes the research foundation for the proposed Walmart sales forecasting project.

Hyndman and Athanasopoulos (2021) explain fundamental time-series forecasting techniques, including exponential smoothing, ARIMA, trend analysis, and seasonality. Their work provides the theoretical foundation for developing traditional forecasting models and evaluating historical retail sales patterns. They also discuss forecasting accuracy measures and model selection, which are useful for comparing alternative forecasting approaches and interpreting prediction errors in this research.

Makridakis et al. (2022) examined the M5 forecasting competition involving Walmart retail sales data. Their study compared forecasting approaches across thousands of sales time series, demonstrating the importance of machine learning, explanatory variables, and forecasting accuracy. The research provides valuable insights into large-scale retail demand prediction and supports the comparison of statistical and machine learning techniques in this project.

Pavlyshenko (2019) investigated machine learning models for sales time-series forecasting, exploring different predictive approaches and model combinations. The study supports the application of machine learning techniques to identify historical sales patterns and

improve forecasting performance. It also highlights the importance of selecting appropriate models and combining predictive information to address complex sales forecasting problems.

Krishna et al. (2018) investigated retail store sales forecasting using regression and boosting techniques. Their findings indicated that boosting algorithms achieved better results than conventional regression approaches, supporting the use of ensemble learning for retail sales prediction. This research provides a useful foundation for exploring advanced predictive algorithms and comparing their effectiveness in forecasting future retail demand.

Sajawal et al. (2022) examined retail sales forecasting using regression and time-series models, including Random Forest, Gradient Boosting, ARIMA, and LSTM. Their findings highlighted XGBoost's predictive performance, demonstrating the potential of ensemble learning for retail demand forecasting. The study also supports evaluating multiple algorithms to determine which approach provides better predictive accuracy for business decision-making.

Ganguly and Mukherjee (2024) compared optimized machine learning algorithms, including Random Forest, Gradient Boosting, SVR, and XGBoost. Their findings demonstrated that hyperparameter optimization significantly improved forecasting performance, highlighting the importance of model tuning in retail sales prediction. This study provides additional support for optimizing model configurations and evaluating different algorithms under consistent experimental conditions.

Eglite and Birzniece (2022) conducted a systematic literature review of deep learning applications in retail sales forecasting. Their research examined different deep learning architectures, evaluation metrics, advantages, and limitations, providing insights into advanced forecasting techniques. Their findings also help explain the growing importance of data-driven

predictive approaches for identifying complex patterns in retail demand and improving forecasting capabilities.

Venkiteela (2025) proposed a machine learning framework for retail sales forecasting involving data preprocessing, exploratory analysis, feature engineering, and regression models. The study reported strong performance from XGBoost, supporting systematic model comparison for retail prediction. It also emphasizes the importance of preparing quality datasets and selecting appropriate predictive variables before implementing forecasting algorithms.

Mejía and Aguilar (2024) developed a hybrid demand forecasting system combining product clustering, feature engineering, and ensemble learning. Their findings demonstrated improved forecasting performance compared with individual models, emphasizing the importance of selecting suitable techniques for different demand patterns. This research highlights how combining data preparation strategies and predictive algorithms can improve forecasting reliability across different product categories.

Swami et al. (2020) investigated retail product sales prediction using XGBoost and Long Short-Term Memory (LSTM) networks. Their experiments showed that XGBoost outperformed LSTM on the selected dataset, highlighting the effectiveness of gradient boosting for retail forecasting. Their work provides useful insights into comparing traditional machine learning and deep learning approaches for predicting future product sales.

The reviewed studies highlight the effectiveness of statistical and machine learning techniques in retail sales forecasting. However, differences in datasets and evaluation methods create opportunities for further investigation. This research addresses this gap by comparing traditional forecasting, Random Forest, and Gradient Boosting using Walmart sales data and consistent evaluation metrics.

VI. METHODOLOGY

A. Research Design

This study will follow a quantitative and experimental research approach to develop and compare machine learning models for retail sales forecasting. The research will use historical Walmart sales data to identify patterns, examine relationships between variables, and predict future sales. The performance of different forecasting techniques will be compared under consistent experimental conditions.

B. Data Collection

The Walmart Store Sales Forecasting dataset will be collected from Kaggle. It includes weekly sales records from 45 stores, along with information about holidays, store characteristics, promotional markdowns, and economic indicators such as fuel prices, CPI, and unemployment rates. The Weekly_Sales variable will serve as the primary dependent variable.

C. Data Preprocessing and Exploratory Analysis

The collected data will be inspected, cleaned, and integrated using Python. Missing values, duplicate records, inconsistent data types, and unusual observations will be examined and addressed appropriately. Exploratory Data Analysis (EDA) will identify sales trends, seasonal variations, holiday effects, and relationships between relevant variables. Charts and statistical summaries will be used to understand the dataset.

D. Feature Engineering and Model Development

Feature engineering will include date-based variables, holiday indicators, historical sales lag values, and moving averages. Four forecasting approaches will be implemented: Seasonal-Naive, Exponential Smoothing or ARIMA, Random Forest Regression, and Gradient Boosting. Hyperparameter tuning will be performed to improve model performance while reducing overfitting.

E. Model Training and Evaluation

The dataset will be divided chronologically into training and testing sets to prevent data leakage. Where appropriate, rolling-origin validation will be applied to examine model stability across different

forecasting periods. Model performance will be compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Weighted Mean Absolute Error (WMAE). Lower error values will indicate better forecasting performance.

F. Software Tools and Result Analysis

Python, Jupyter Notebook, Pandas, NumPy, Scikit-learn, Statsmodels, Matplotlib, and Seaborn will support implementation, analysis, and visualization. Finally, actual and predicted sales will be compared using charts and performance tables. Feature importance and forecasting errors will also be examined to identify influential variables and limitations. The findings will be used to recommend the most reliable forecasting approach for supporting retail inventory management and business planning.

VII. POTENTIAL CHALLENGES AND MITIGATION STRATEGIES

Potential challenges include missing values, seasonal sales fluctuations, model overfitting, and data leakage. Missing values will be handled through appropriate data-cleaning techniques. Holiday indicators and time-based features will help capture seasonal variations. Chronological data splitting and hyperparameter tuning will reduce data leakage and overfitting. Computational challenges will be managed by using efficient algorithms and optimized model configurations.

VIII. EXPECTED OUTCOMES AND DELIVERABLES

This research is expected to identify an effective machine learning approach for retail sales forecasting. The study will identify important sales trends, seasonal patterns, and factors influencing retail demand. The main deliverables will include a clean dataset, exploratory data analysis, developed forecasting models, performance comparison tables, and visualizations. A reproducible Python notebook, detailed research report, practical recommendations, and final presentation will also be prepared.

IX. SCOPE AND LIMITATIONS

This research will focus on weekly retail sales forecasting using the Walmart Store Sales Forecasting dataset from Kaggle. It will analyze historical sales, holiday indicators, promotional markdowns, store characteristics, and economic variables. The study will compare traditional statistical methods with selected machine learning algorithms. However, the dataset covers a limited historical period and represents only Walmart stores in the United States. Missing promotional information and changing market conditions may affect prediction accuracy. The findings may therefore not be directly applicable to all retail businesses. The project will focus on model development and evaluation rather than implementing a real-time forecasting system.

X. PROJECT ACTIVITIES AND TIMELINE

Here is the detailed project timeline:

TABLE 1

PROJECT TIMELINE

Week	Deadline	Project Activities and Task	Expected Deliverables
Week 1	October 25, 2026	Finalize the research topic, define the problem and objectives, identify the Walmart dataset, conduct preliminary research, and submit the short proposal.	Short Proposal
Week 2	November 1, 2026	Review at least 10 academic sources, finalize research objectives and methodology, identify software tools, and submit the full research proposal.	Full Research Proposal
Week 3	November 8, 2026	Download the Walmart dataset, examine its structure, identify missing values and duplicates, integrate relevant files, and complete data preprocessing.	Project Update #1 and Prepared Dataset
Week 4	November 15, 2026	Perform exploratory data analysis, examine sales trends, seasonal variations, and holiday effects, analyze relevant variables, and prepare visualizations.	Project Update #2 and EDA Results

Week 5	November 22, 2026	Perform feature engineering, create historical sales variables, develop the seasonal-naive baseline model, and schedule the oral presentation.	Project Update #3 and Baseline Model
Week 6	November 29, 2026	Develop statistical forecasting, Random Forest, and Gradient Boosting models, perform hyperparameter tuning, and generate preliminary predictions.	Project Update #4 and Developed Models
Week 7	December 6, 2026	Evaluate forecasting models using MAE, RMSE, and WMAE, compare results, analyze errors, identify the best model, and submit the draft report.	Project Report Draft
Week 8	December 13, 2026	Incorporate feedback, finalize research findings, complete the final report, prepare presentation slides, and deliver the oral presentation.	Final Project Report and Oral Presentation

XI. CONCLUSION

This research proposes the development and evaluation of machine learning models for predicting retail sales using the Walmart Store Sales Forecasting dataset. By analyzing historical sales patterns, seasonal variations, holidays, and economic factors, the study aims to improve forecasting accuracy. Different statistical and machine learning techniques will be compared using appropriate evaluation metrics to identify the most reliable approach. The findings are expected to demonstrate the potential of machine learning in retail demand prediction and provide practical insights for inventory management, resource allocation, and business planning. Ultimately, the project aims to support more informed and data-driven retail business decisions.

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